

AI Calculator · Multiple Choice Test

Scan one problem at a time. Each should come back as CHOICE + the option label. Answer key on the last page (keep it out of the camera's view).

1. If $3x - 5 = 16$, then $x =$

- A) 3 B) 7 C) $11/3$ D) 21
-

2. $d/dx [\sin(x^2)] =$

- A) $\cos(x^2)$
B) $2x \cos(x^2)$
C) $2x \sin(x^2)$
D) $-2x \cos(x^2)$
-

3. $\int_0^{\pi} \sin x \, dx =$

- A) 0 B) 1 C) 2 D) π
-

4. $\lim_{x \rightarrow 0} \frac{\sin(3x)}{x} =$

- A) 0 B) 1 C) 3 D) does not exist
-

5. A 2.0 kg object accelerates at $3.0 \, \text{m/s}^2$. What is the net force on it?

- (a) 0.67 N (b) 1.5 N (c) 5.0 N (d) 6.0 N (e) 9.0 N
-

6. Which sample contains the most moles?

- 1) 10 g of H_2
2) 10 g of He
3) 10 g of O_2
4) 10 g of CO_2
-

7. What is the pH of 0.001 M HCl?

- A) 1 B) 2 C) 3 D) 11
-

8. Which organelle produces most of a cell's ATP?

- A) Ribosome
B) Mitochondrion
C) Golgi apparatus
D) Lysosome
-

9. The angle between $\mathbf{u} = \langle 1, 0, 1 \rangle$ and $\mathbf{v} = \langle 0, 1, 1 \rangle$ is

- A) 30° B) 45° C) 60° D) 90°
-

10. $\det \begin{vmatrix} 2 & 1 \\ 5 & 3 \end{vmatrix} =$

- A) 1 B) 11 C) -1 D) 6
-

11. A ball is dropped from rest 20 m above the ground. About how long until it lands?
($g = 9.8 \text{ m/s}^2$)

- A) 1.0 s B) 1.5 s C) 2.0 s D) 2.5 s
-

12. Which is the solution set of $x^2 + 4 = 0$?

- A) $x = 2$
B) $x = -2$
C) $x = \pm 2i$
D) no solution
-

Handwritten round

Copy each into its space by hand, including the options. Same scan rules.

H1. Copy #1 with its A–D options

H2. Copy #4 (the limit) with its options

H3. Copy #6 with its 1–4 options, in pencil

H4. Write by hand: $5 \times 7 - 3 = ?$ Write by hand: $5 \times 7 - 3 = ?$ A) 32 B) 20 C) 38 D) 35
A) 32 Write by hand: $5 \times 7 - 3 = ?$ A) 32 B) 20 C) 38 D) 35
B) 20 C) 38 D) 35

Answer key and score sheet

Result: ✓ right letter · G = right letter shown as GUESS · ✗ wrong letter · N = no CHOICE shown (answered without picking an option).

#	Correct choice	Result	% sure	Notes / what it misread
1	B $x = 7$			
2	B $2x \cos(x^2)$, chain rule			
3	C 2			
4	C 3			
5	(d) 6.0 N, $F = ma$			
6	1 10 g $H_2 \approx 4.96$ mol (He 2.50, O_2 0.31, CO_2 0.23)			
7	C pH 3			
8	B Mitochondrion			
9	C 60° : $\cos \theta = 1/(\sqrt{2} \cdot \sqrt{2}) = \frac{1}{2}$			
10	A $1: 2 \cdot 3 - 1 \cdot 5$			
11	C $2.02 \text{ s} \approx 2.0 \text{ s}$ (closest option)			
12	C $x = \pm 2i$ (D if it assumes real numbers only: note which)			
H1	B			
H2	C			
H3	1			
H4	A 32			

- This sheet tests the option labels in four styles: A–D in a row, A–D stacked, (a)–(e), and 1–4.
- #11's exact answer (2.02 s) isn't an option: it should pick C and may note the rounding.
- Any "N" result means it missed that the problem had options. Tell me which ones.